Supporting Information for "Attack-Method Choice Changes Robustness Conclusions for a Standard LSTM Rainfall-Runoff Model"

Y. Sun[1]

[1]College of Hydrology and Water Resources, Hohai University, Nanjing, China

Contents

Text S1. Model Training Configuration

The LSTM model was trained using the neuralhydrology framework with the configuration summarized in Table S1. The model uses five dynamic Daymet forcing variables and 14 static catchment attributes. Training was conducted for 30 epochs with a single-layer LSTM (128 hidden units), AdamW optimization, and NSE loss.

The adversarial study uses the local 531-basin temporal benchmark run stored under `results/05_adversarial_robustness/runs/reproduce_531_nse074_2025_1129_2145_ep30/`. The checkpoint is pinned to `model_epoch020.pt`; later saved checkpoints were not used for adversarial evaluation.

Table 1 — Model training hyperparameters.

Parameter	Value
Architecture	LSTM (single-layer)
Hidden units	128
Training epochs	30
Batch size	256
Sequence length	365 days
Optimizer	AdamW
Loss function	NSE
Dynamic inputs	Precipitation, $T_{\max}$, $T_{\min}$, shortwave radiation, vapor pressure
Static attributes	14 CAMELS attributes
Training period	01/10/1990–30/09/1995
Validation period	01/10/1995–30/09/2000
Test period	01/10/2000–30/09/2005
Random seed	159193

Text S2. Basin Coverage and Experiment Counts

The adversarial rerun uses the same 531 basin IDs as the temporal benchmark model. The same basin list is used for all adversarial experiment blocks; there is no separate basin-list merge from the previous result line.

After merging and deduplication, the full evaluation contains 17,523 individual experiment records across 531 unique basin IDs. Table S2 summarizes basin counts by experiment block.

Table 2 — Basin counts per experiment block.

Experiment block	Description	Catchments
Exp. 1	Attack comparison (5 methods $\times$ 4 ε values)	531
Exp. 2	Constraint ablation (3 constraints $\times$ 3 ε values)	531
Exp. 3	Targeted attacks (3 targets at $\varepsilon = 0.1$)	531
Exp. 4	Causal trigger (4 pre-event windows)	531
Exp. 5	C&W minimum perturbation	531
Merged total	Unique basin IDs across all experiment records	531
Merged total	Individual experiment records	17,523

Text S3. Finite NSE Metric Handling

All experiment cells cover the 531 benchmark basin IDs. However, the Nash–Sutcliffe Efficiency is undefined or numerically unstable for evaluation windows that contain too few valid observations or have near-constant observed discharge, because NSE divides by the variance of the observed flow over the window: as that variance approaches zero, the metric becomes arbitrarily sensitive to even sub-millimetre changes in modelled flow. To avoid these degenerate cases, the adversarial runner selects evaluation windows that contain more than 100 valid daily observations and whose observed-flow standard deviation, normalized to the basin's clean-flow scale, exceeds 0.1; this rules out windows that are essentially flat or that contain only a single short discharge event. Seventeen of the 531 basin IDs did not satisfy this variability requirement in any candidate window, and therefore yield NaN values for $\mathrm{NSE}_{\mathrm{clean}}$, $\mathrm{NSE}_{\mathrm{adv}}$, and ΔNSE rather than meaningful robustness numbers.

The affected basin IDs are: 04127997, 05120500, 06350000, 06404000, 06406000, 06409000, 06431500, 06447500, 06847900, 07142300, 07299670, 07301410, 08324000, 09306242, 09386900, 09447800, and 09484600. Raw streamflow and Daymet forcing files exist for these basins, and the test-period raw streamflow records are complete; the NaN values arise from the variance-based finite-metric guard rather than from missing data, and these basins remain part of the 531-basin coverage statement everywhere experiment provenance is documented. Main-text NSE headline statistics use only the 514 basins with finite metrics so that medians and IQRs are not contaminated by undefined values.

References

Kratzert, F., Gauch, M., Nearing, G., & Klotz, D. (2022). NeuralHydrology—A Python library for deep learning research in hydrology. *Journal of Open Source Software*, *7*(71), 4050.

Table 3 — Rerun provenance artifacts.

Artifact	Path or value
Victim run	`results/05_adversarial_robustness/runs/reproduce_531_nse074_2025_1129_2145_ep30/`
Pinned checkpoint	`model_epoch020.pt`
Adversarial config	`src/adversarial/configs/full_eval_531_epoch020.yaml`
Basin list	`src/adversarial/data/531_basins.txt`
Output root	`results/adversarial_eval/531_epoch020/`
Merged result file	`results/adversarial_eval/531_epoch020/merged_all.json`
Coverage audit	17,523 records; 531 unique basin IDs; no missing basin IDs
Finite NSE metrics	514 basins; 17 low-variance-window NaN basins